

Scene Setup Guide — Space Shooter

This document provides exact, step-by-step instructions for setting up each scene in the Unity Editor.

Scene 1: MainMenu

Hierarchy Structure

MainMenu (Scene)	
Main Camera	
GameManager	Empty GO , GameManager.cs, DontDestroyOnLoad
AudioManager	Empty GO , AudioManager.cs + 2 AudioSources, DontDestroyOn-
Load	
Canvas	
TitleText	UI > Text
HighScoreText	UI > Text
PlayButton	UI > Button
QuitButton	UI > Button
VersionText	UI > Text
EventSystem	Auto-created with Canvas

Camera Setup

- **Clear Flags:** Solid Color
- **Background:** #050510 (very dark blue-black)
- **Size:** 5 (default orthographic)

GameManager GameObject

1. Create: **GameObject** → **Create Empty**, name `GameManager`
2. Add Component: `GameManager` script
3. Leave Player and EnemySpawner references empty (they exist only in Gameplay scene)

AudioManager GameObject

1. Create: **GameObject** → **Create Empty**, name `AudioManager`
2. Add Component: `AudioManager` script
3. Add Component: `AudioSource` (for music)
 - Loop: ✓, Play On Awake: ✗
4. Add Component: `AudioSource` (for SFX)
 - Loop: ✗, Play On Awake: ✗
5. In AudioManager Inspector:
 - Drag music AudioSource → **Music Source**
 - Drag SFX AudioSource → **SFX Source**
 - Assign audio clips as available

Canvas Setup

1. Create: **GameObject** → **UI** → **Canvas**

2. Canvas component:

- Render Mode: Screen Space - Overlay

3. Canvas Scaler:

- UI Scale Mode: Scale With Screen Size
- Reference Resolution: 800 × 600
- Match: 0.5

4. Add Component: MenuManager script to Canvas

UI Elements (children of Canvas)

TitleText

- Create: Right-click Canvas → **UI** → **Text**
- Name: TitleText
- Rect Transform: Anchor to top-center
- Pos Y: 150 , Width: 600 , Height: 80
- Text: SPACE SHOOTER
- Font Size: 48
- Alignment: Center/Middle
- Color: White
- **Assign to** MenuManager → Title Text

HighScoreText

- Create: **UI** → **Text**, name HighScoreText
- Pos Y: 50 , Width: 400 , Height: 40
- Font Size: 24 , Color: #FFFF00 (Yellow)
- Text: High Score: 0
- **Assign to** MenuManager → High Score Text

PlayButton

- Create: **UI** → **Button**, name PlayButton
- Pos Y: -50 , Width: 200 , Height: 50
- Child Text: PLAY , Font Size: 28
- Button colors: Normal=Dark Blue, Highlighted=Light Blue
- **Assign to** MenuManager → Play Button

QuitButton

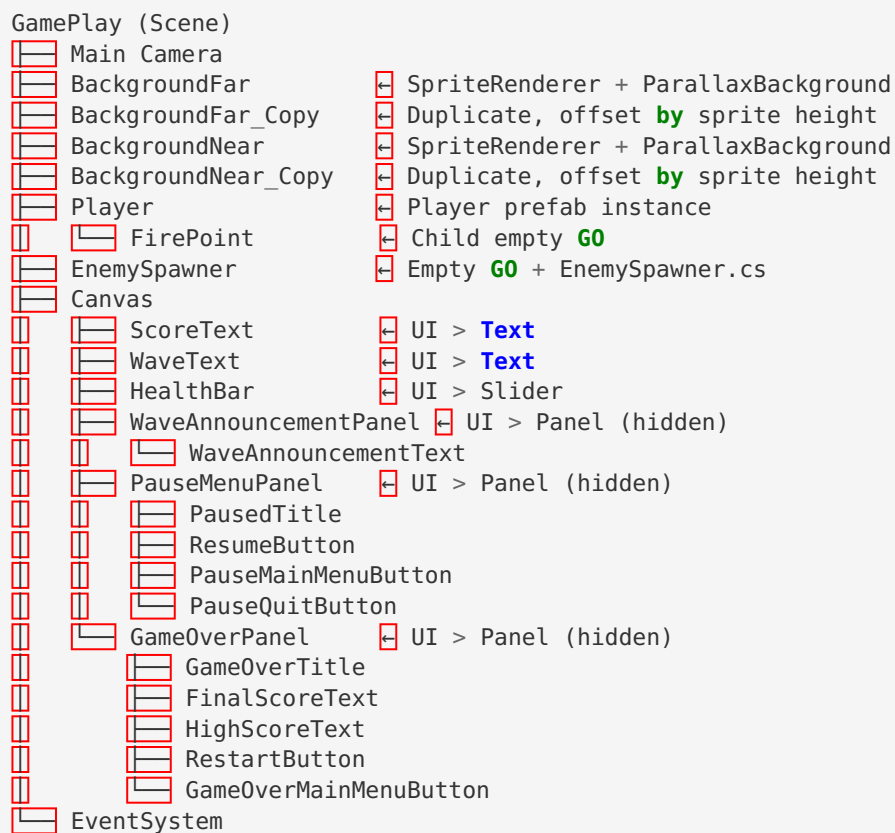
- Create: **UI** → **Button**, name QuitButton
- Pos Y: -120 , Width: 200 , Height: 50
- Child Text: QUIT , Font Size: 28
- **Assign to** MenuManager → Quit Button

VersionText

- Create: **UI** → **Text**, name VersionText
 - Anchor: Bottom-Right, Font Size: 14 , Color: Gray
 - **Assign to** MenuManager → Version Text
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Scene 2: Gameplay

Hierarchy Structure



Camera Setup

- Same as MainMenu: Solid Color, #050510, Size 5

Background Layers

BackgroundFar

1. Create: **GameObject** → **Create Empty**, name BackgroundFar
2. Add: **SpriteRenderer**
 - Sprite: BackgroundStarsFar
 - Sorting Layer: Background, Order: 0
 - Draw Mode: Simple
3. Add: **ParallaxBackground** script
 - Scroll Speed: 0.5
 - Sprite Height: 10.24 (1024px ÷ 100 PPU)
4. Position: (0, 0, 1) (Z=1 to be behind gameplay)

BackgroundFar_Copy

1. Duplicate BackgroundFar (Ctrl+D)
2. Move to: (0, 10.24, 1) — exactly one sprite height above

BackgroundNear

1. Create another pair with BackgroundStarsNear sprite
2. Sorting Layer: Background, Order: 1

3. Scroll Speed: 1.5
4. Position: (0, 0, 0.5)
5. Duplicate and offset same as above

Player Setup

1. Drag Player prefab from Assets/Prefabs/ into scene
2. Position: (0, -3, 0)
3. Verify all PlayerController references are set

EnemySpawner

1. Create: **GameObject** → **Create Empty**, name EnemySpawner
2. Add: **EnemySpawner** script
3. Assign in Inspector:
 - Basic Enemy Prefab → EnemyBasic
 - Zigzag Enemy Prefab → EnemyZigzag
 - Heavy Enemy Prefab → EnemyHeavy

GamePlay Canvas

1. Create Canvas (same settings as MainMenu)
2. Add: **UIManager** script to Canvas

ScoreText

- Anchor: Top-Left
- Pos: (120, -30), Width: 200, Height: 30
- Text: Score: 0, Font Size: 22, Color: White
- **Assign to** UIManager → Score Text

WaveText

- Anchor: Top-Center
- Pos: (0, -30), Width: 200, Height: 30
- Text: Wave: 1, Font Size: 22, Color: White
- **Assign to** UIManager → Wave Text

HealthBar (Slider)

- Create: **UI** → **Slider**, name HealthBar
- Anchor: Top-Right
- Pos: (-120, -30), Width: 200, Height: 20
- **Uncheck** Interactable on Slider component
- Min Value: 0, Max Value: 100, Value: 100
- Background: Set Image color to #330000 (dark red)
- Fill Area → Fill: Set Image color to #00FF00 (green)
- Remove Handle Slide Area (delete the child)
- **Assign Slider to** UIManager → Health Bar
- **Assign Fill Image to** UIManager → Health Bar Fill

WaveAnnouncementPanel

- Create: **UI** → **Panel**, name WaveAnnouncementPanel
- Anchor: Stretch to full screen
- Image color: rgba(0, 0, 0, 0) (fully transparent)

- Child Text `WaveAnnouncementText` :
- Center of screen, Font Size: `48` , Color: `#FFFF00`
- Text: `WAVE 1`
- **Set panel inactive** in Inspector (uncheck checkbox)
- **Assign to** UIManager → Wave Announcement Panel + Text

PauseMenuPanel

- Create: **UI → Panel**, name `PauseMenuPanel`
- Stretch to full screen
- Image color: `rgba(0, 0, 0, 180)` (semi-transparent black)
- Children:
- Text `PausedTitle` : “PAUSED”, Font Size `42` , White, centered at `(0, 100)`
- Button `ResumeButton` : “RESUME”, `(0, 0)`
- Button `PauseMainMenuButton` : “MAIN MENU”, `(0, -70)`
- Button `PauseQuitButton` : “QUIT”, `(0, -140)`
- **Set panel inactive**
- **Assign all references** to UIManager

GameOverPanel

- Create: **UI → Panel**, name `GameOverPanel`
- Stretch to full screen
- Image color: `rgba(0, 0, 0, 200)`
- Children:
- Text `GameOverTitle` : “GAME OVER”, Font Size `48` , Red, `(0, 120)`
- Text `FinalScoreText` : “Score: 0”, Font Size `28` , White, `(0, 50)`
- Text `HighScoreText` : “High Score: 0”, Font Size `22` , Yellow, `(0, 10)`
- Button `RestartButton` : “RESTART”, `(0, -60)`
- Button `GameOverMainMenuButton` : “MAIN MENU”, `(0, -130)`
- **Set panel inactive**
- **Assign all references** to UIManager

Build Settings Final Checklist

1. **File → Build Settings**
2. Scenes in build:
 - ☒ `Assets/Scenes/MainMenu.unity` — Index 0
 - ☒ `Assets/Scenes/GameOver.unity` — Index 1
3. Platform: `PC, Mac & Linux Standalone`
4. Target: `Windows` , Architecture: `x86_64`
5. Player Settings:
 - Product Name: `Space Shooter`
 - Default Resolution: `800 × 600`
 - Fullscreen Mode: `Windowed`

Prefab Reference Checklist

Prefab	Script	Key Fields to Assign
Player	PlayerController	bulletPrefab → PlayerBullet, firePoint → FirePoint child
EnemyBasic	EnemyController	powerUpPrefabs → [PowerUp- Weapon, PowerUpShield, PowerUpHealth]
EnemyZigzag	EnemyController	powerUpPrefabs → [PowerUp- Weapon, PowerUpShield, PowerUpHealth]
EnemyHeavy	EnemyController	enemyBulletPrefab → Enemy- Bullet, powerUpPrefabs → [...]
PlayerBullet	BulletController	damage → 10
EnemyBullet	BulletController	damage → 15
PowerUpWeapon	PowerUpController	powerUpType → WeaponUp- grade
PowerUpShield	PowerUpController	powerUpType → Shield
PowerUpHealth	PowerUpController	powerUpType → Heal- thRestore

Tag Assignment Checklist

GameObject/Prefab	Tag
Player	Player
EnemyBasic	Enemy
EnemyZigzag	Enemy
EnemyHeavy	Enemy
PlayerBullet	PlayerBullet
EnemyBullet	EnemyBullet
PowerUpWeapon	PowerUp
PowerUpShield	PowerUp
PowerUpHealth	PowerUp